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SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks: 30

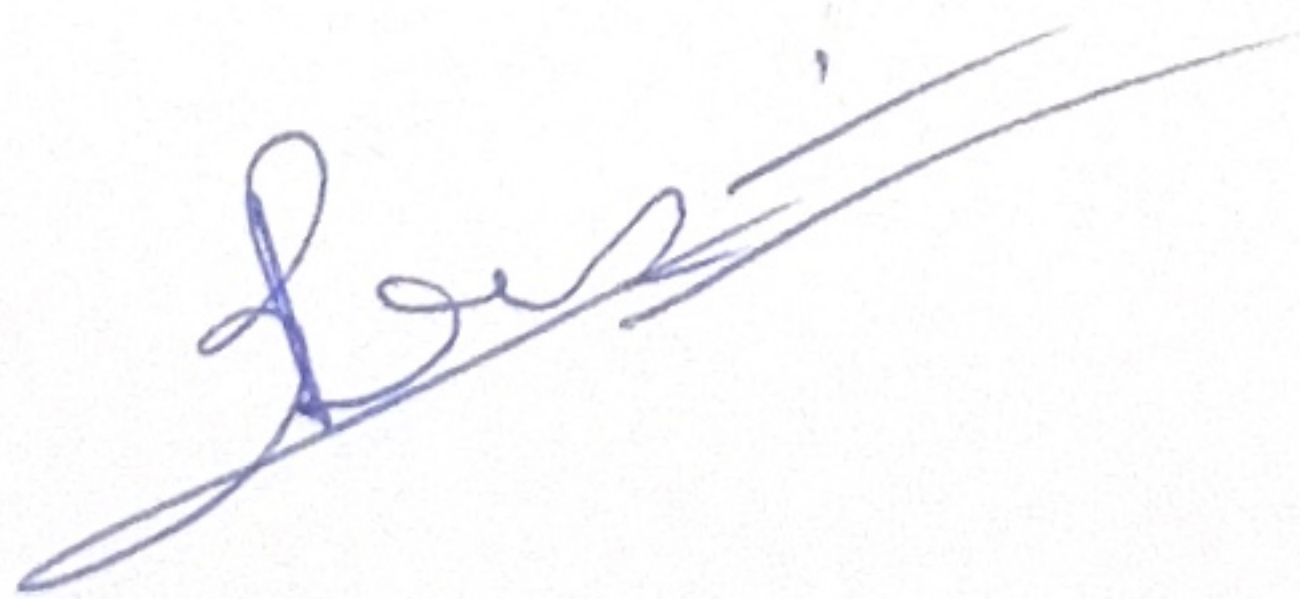
Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M. PRASANNA VENKATESH (192573016) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Accepted 4



Student 16: M Prasanna Venkatesh | Register Number: 192573016

1. A college must assign 4 employees (E1, E2, E3, E4) to 4 administrative tasks (Admissions, Accounts, Records, IT Support) such that:

Constraints:

- i. E1 can only handle Admissions or Accounts
- ii. E2 must be assigned before E3 (Admissions < Accounts < Records < IT Support)
- iii. Only one employee per task
- iv. E4 cannot handle Records
- v. All employees must be assigned exactly one task

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule.

2. A security patrol robot operates in a 6x6 building-floor grid from Start (S) to a checkpoint (G). Certain zones are restricted (no-entry) and some corridors carry extra monitoring cost.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Normal move cost = 1, monitored-corridor move cost = 2
- iii. Cannot pass through restricted (no-entry) cells
- iv. The robot must minimize total path cost

Using the above constraints and Uniform Cost Search (UCS), show step-by-step node expansion and priority-queue updates, and determine the optimal path and cost.



Backtracking Search: Employee - Task Assignment

Problem: Assign employees E_1, E_2, E_3, E_4 to tasks Admissions, Accounts, Records, IT support (on each task) satisfying all given constraints.

Constraints.

- * $C_1: E_1 \in \{\text{Admissions, Accounts}\}$
- * $C_2: \text{Task}(E_2)$ must precede $\text{Task}(E_3)$ in the order $\text{Admission}(1) < \text{Accounts}(2) < \text{Records}(3) < \text{IT support}(4)$
- * $C_3: \text{Each task assigned to exactly one employee (one to one)}$
- * $C_4: E_4 \neq \text{Records}$
- * $C_5: \text{All four employees must be Assigned}$

Backtracking search trace (Variables assigned in order $E_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$; domain values tried in order Admissions, Accounts, Records, IT support).

step	variable	value Tried	constraint check	Result.
1.	E_1	Admissions	$C_1: \text{Admissions} \in \{\text{Admission, Accounts}\} \rightarrow$ Satisfied.	Accept, Assign $E_1 = \text{Admissions}$

2.	E ₂	Accounts	No conflict yet (C ₂ needs E ₃ to verify.) → provisionally OK	Accept, Assign E ₂ = Accounts
3.	E ₃	Records	C ₂ : Task(E ₂) = Accounts(2) C ₃ : Task(E ₃) = Records(3) → satisfied.	Accept, Assign E ₃ = Records
4.	E ₄	IT support	C ₄ : IT support ≠ Records → satisfied; only value left in domain	Accept, Assign E ₄ = IT support

All four variables are now assigned with no constraint violated at any depth, so that search terminates successfully at the first branch - no backtracking. (dead-end) was required in this trace, since Admissions / Accounts / Records / IT support were tried in constraint-consistent order.

Final valid schedule.

Employee · Assigned Task.

E₁ · Admissions.

E₂ · Accounts.

E₃ · Records.

E₄ · IT support.

Uniform Cost search: security Patrol Robot.

Assumed Grid Layout (rows 1-6 top-bottom, columns 1-6, left → right)

* start $s = (1, 1)$; Goal $g = (4, 4)$

* Restricted (no-entry) cells: $(2, 2)$ $(3, 2)$ $(2, 3)$

* Monitored corridor. (cost = 2 per move): column 4 rows 1-3 → $(1, 4)$, $(2, 4)$ $(3, 4)$ and row 4 columns 1-3 → $(4, 1)$ $(4, 2)$ $(4, 3)$.

* All other moves cost = 1.

UCS Node Expansion (priority queue ordered by cumulative path cost g)

step	Node Expanded	g (cost so far)	children Generated (cell: g)	Queue after step (sorted)
1.	$(1, 1)$	0	$(1, 2): 1$, $(2, 1): 1$	$(1, 2): 1$, $(2, 1): 1$
2.	$(1, 2)$	1	$(1, 3): 2$, $\{(2, 2) \text{ blocked}\}$	$(2, 1): 1$, $(1, 3): 2$
3.	$(2, 1)$	1	$(3, 1): 2$ $\{(2, 2), \text{ blocked}\}$	$(1, 3): 2$, $(3, 1): 2$
4.	$(1, 3)$	2	$(1, 4): 4$ {monitored} $\{(2, 3) \text{ blocked}\}$	$(3, 1): 2$, $(1, 4): 4$
5.	$(3, 1)$	2	$(4, 1): 4$ {monitored}	$(1, 4): 4$, $(4, 1): 4$
6.	$(1, 4)$	4	$(2, 4): 6$ {monitored}	$(4, 1): 4$, $(2, 4): 6$
7.	$(4, 1)$	4	$(4, 2): 6$ {monitored}	$(2, 4): 6$, $(4, 2): 6$

8	(2, 4)	6	(3, 4): 8 {monitored}	(4, 2): 6, (3, 4): 8
9	(4, 2)	6	(4, 3): 8 {monitored}	(3, 4): 8, (4, 3): 8
10	(3, 4)	8	(4, 4): 9 ← Goal reached cost 9	(4, 3): 8, (4, 4): 9
11	(4, 3)	8	(4, 4): 9 ← GOAL, tie same cost 9	(4, 4): 9 (goal-expand next)
12	(4, 4)	9	GOAL popped from queue → search terminates	-

UCS always expands the lowest-g node from the priority queue, so the goal is guaranteed optimal once popped. Two equal-cost optimal paths exist because the grid is symmetric about the diagonal between the two monitored strips.

Optimal Path and Cost:

Path A: (1, 1) → (1, 2) → (1, 3) → (1, 4) → (2, 4) → (3, 4) → (4, 4)

Path B: (1, 1) → (2, 1) → (3, 1) → (4, 1) → (4, 2) → (4, 3) → (4, 4)

Total optimal cost = 9 units. using either path - 4 normal moves ... reconciled as: Path A moves =

1(→ 1, 2) + 1(→ 1, 3) + 2(→ 1, 4) + 2(→ 2, 4) + 2(→ 3, 4) + 1(→ 4, 4)
 = 9. Both paths avoid all restricted cells (2, 2), (3, 2), (2, 3) and satisfy every constraint.

RUBRICS FOR CALCULATION:

Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

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